

Solutions — Problem Set: Unconfoundedness

BUSS975 — covering Lectures 3a and 3b

Solution key. Every numerical result below is produced by the code shown — the document recomputes it at render time, so the printed numbers and the code cannot drift apart.

For the written parts, what follows is a model answer, not the only acceptable one. Grade on whether the mechanism is identified correctly, not on wording.

Problem 1 (Covariates, estimands, and the two assumptions)

(a) Classifying the variables

Variable	Type	For the total effect
(i) size, industry, rating	Confounder	Include. They drive both the bank's decision and investment; conditioning closes the backdoor paths.
(ii) prior-year export demand	Precision covariate	Optional; include. It predicts Y but not D , so it cannot bias anything and it shrinks the standard error. It is pre-treatment, so it cannot be an outcome.
(iii) amount drawn	Mediator	Exclude. It lies on $D \rightarrow \text{drawing} \rightarrow Y$; conditioning on it removes part of the effect you want.
(iv) later analyst coverage	Collider	Exclude. $D \rightarrow \text{coverage} \leftarrow Y$ is closed; conditioning on it opens it. It is also post-treatment.
(v) managerial quality	Unobserved confounder	Cannot be included. This is the threat to the design.

The variable whose treatment depends on the question is **(iii), the amount drawn**. Conditioning on it answers a *partial-effect* question: *did the credit line raise investment through channels other than the funds actually drawn* — for instance as precautionary liquidity that lets the firm commit to projects — holding fixed how much it drew? One caveat worth stating: even that partial effect is identified only if nothing unobserved drives both drawing and investment, and managerial quality (v) is an obvious candidate.

(b) Two questions, two estimands

The regulator's question — every firm with a line — needs the *ATE*. The lending committee's question — the firms that got one — needs the *ATT*.

- The *ATE* requires **strong unconfoundedness**, $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$, and **strong common support**, $0 < e(X) < 1$ everywhere.
- The *ATT* requires only **weak unconfoundedness**, $Y^0 \perp\!\!\!\perp D \mid X$, and **weak common support**, $e(X) < 1$ wherever treated firms are.

The *ATT* gets away with less because for the treated firms Y^1 is *observed*: only Y^0 is missing, and it is imputed from controls in the treated firms' own cells. Nothing is ever imputed for a control.

(c) Identifying the ATT

$$\begin{aligned} ATT(x) &= E[Y^1 \mid D=1, X=x] - E[Y^0 \mid D=1, X=x] \\ &= E[Y \mid D=1, X=x] - E[Y^0 \mid D=1, X=x] && \text{switching equation } (Y = Y^1 \text{ when } D=1) \\ &= E[Y \mid D=1, X=x] - E[Y^0 \mid D=0, X=x] && \text{weak unconfoundedness, } Y^0 \perp\!\!\!\perp D \mid X \\ &= E[Y \mid D=1, X=x] - E[Y \mid D=0, X=x] && \text{switching equation } (Y = Y^0 \text{ when } D=0) \end{aligned}$$

Unconfoundedness is used **once, on Y^0 , in the second (missing) term** — never on Y^1 . The *ATT* then averages $ATT(x)$ with weights $\Pr(X=x \mid D=1)$, the covariate distribution among the treated. Cells with no treated firms get zero weight, which is why the *ATT* needs controls only where treated firms are.

(d) Imbens's third defence

The defence: the decision-maker optimises something other than your outcome. A bank prices a credit line to manage default risk, relationship value and capital charges; a firm seeks one as liquidity insurance and covenant slack. Neither is choosing on next year's capital expenditure. Conditional on the credit-relevant observables — size, industry, rating — the residual variation in who gets a line is then plausibly unrelated to the firm's investment potential outcomes.

It collapses **when an agent optimises precisely your outcome**: if firms apply for lines *because* they have specific projects lined up, or banks grant them on the strength of management's investment plans, then D is chosen on $Y^1 - Y^0$ directly. Managerial quality (v) is exactly such a channel if good managers both secure lines and invest better.

(e) NSW: which assumption was each paper about?

Dehejia and Wahba were about **common support**. Their gain came from discarding comparison units from the CPS and PSID that were utterly unlike NSW participants — imposing overlap. Much of LaLonde's failure had been extrapolation, not confounding.

Smith and Todd were about **unconfoundedness**, and the specification choices that stand in for it. The recovery was sensitive to the subsample and to the score specification, so whether the

observables really make assignment ignorable was never settled — and the specification that worked may have been found because the benchmark was known.

You cannot use Dehejia and Wahba as a check because **they had an experimental benchmark to grade against, and you do not** — that is why you are doing the study. You can check overlap with a histogram of $\hat{e}$. The part Smith and Todd showed to be fragile, unconfoundedness, cannot be checked in your data at all.

Problem 2 (Subclassification, estimands, and common support)

```
N <- c(A = 400, B = 400, C = 200)
nT <- c(A = 100, B = 200, C = 180)
nC <- N - nT
y1 <- c(A = 14, B = 20, C = 31)      # E[Y | D=1, X]
y0 <- c(A = 10, B = 18, C = 25)      # E[Y | D=0, X]
e <- nT / N
tau <- y1 - y0
rbind(e = e, tau = tau)
```

```
      A    B    C
e  0.25 0.5 0.9
tau 4.00 2.0 6.0
```

(a) Three estimands

```
ATE <- sum(N * tau) / sum(N)      # weight: population share
ATT <- sum(nT * tau) / sum(nT)    # weight: treated share
ATU <- sum(nC * tau) / sum(nC)    # weight: control share
pi_ <- sum(nT) / sum(N)
round(c(ATE = ATE, ATT = ATT, ATU = ATU, pi = pi_,
        check = pi_ * ATT + (1 - pi_) * ATU), 4)
```

```
      ATE    ATT    ATU    pi  check
3.6000 3.9167 3.3077 0.4800 3.6000
```

The same three within-stratum effects, (4, 2, 6), weighted three ways: $ATE = 3.60$, $ATT = 3.92$, $ATU = 3.31$, and $0.48 \times 3.92 + 0.52 \times 3.31 = 3.60$.

(b) The SDO and its decomposition

```

EY_T    <- sum(nT * y1) / sum(nT)           # E[Y | D=1]
EY_C    <- sum(nC * y0) / sum(nC)           # E[Y | D=0]
SDO     <- EY_T - EY_C
levels  <- sum(nT * y0) / sum(nT) - EY_C     # E[Y0|D=1] - E[Y0|D=0]
gains   <- (1 - pi_) * (ATT - ATU)
round(c(SDO = SDO, ATE = ATE, levels = levels, gains = gains,
        total = ATE + levels + gains), 4)

```

```

      SDO    ATE levels  gains  total
9.2212 3.6000 5.3045 0.3167 9.2212

```

$SDO = 9.22$, more than two and a half times the ATE . Of the gap of 5.62, **selection on levels contributes** 5.30 and selection on gains only 0.32. Levels dominate because the waivers went disproportionately to stratum C, whose banks grow fast even without a waiver ($Y^0 = 25$).

(c) Why the ATT exceeds the ATE

The regulator concentrated waivers in stratum C ($e = 0.9$), which also has the largest effect (6); the ATT weights by the treated distribution, so it over-weights exactly the stratum where the waiver works best.

(d) Stratum A with no treated banks

```

ATT_noA <- sum(nT[c("B", "C")] * tau[c("B", "C")]) / sum(nT[c("B", "C")])
ATT_noA

```

```
[1] 3.894737
```

Only the ATT survives, at 3.89: it weights strata by the treated distribution, and A would have no treated banks, so it gets zero weight. The ATE and ATU both need $E[Y | D=1, X=A]$, which would no longer exist.

(e) Stratum C with no controls

```

ATU_noC <- sum(nC[c("A", "B")] * tau[c("A", "B")]) / sum(nC[c("A", "B")])
ATU_noC

```

```
[1] 3.2
```

Only the *ATU* survives, at 3.20. The *ATE* and *ATT* both need $E[Y \mid D=0, X=C]$.

The rule: an estimand survives if it puts zero weight on the empty cell. The *ATT* needs controls wherever there are treated units (weak common support); the *ATU*, symmetrically, needs treated units wherever there are controls; the *ATE* needs both everywhere.

With ten binary covariates there are $2^{10} = 1,024$ cells. A sample of a few thousand leaves many of them with no treated units or no controls, so empty cells are the **norm** — the curse of dimensionality.

Problem 3 (Matching: the discrepancy, the correction, the standard error)

```
xT <- c(2, 6, 8, 10); yT <- c(30, 44, 52, 58)      # treated
xC <- c(1, 4, 9, 30); yC <- c(10, 16, 26, 0)      # controls
m   <- sapply(xT, function(x) which.min(abs(xC - x))) # index of each match
tauh <- yT - yC[m]
data.frame(X_i = xT, match_X = xC[m], Y_i = yT, Y_match = yC[m], tau_i = tauh)
```

	X_i	match_X	Y_i	Y_match	tau_i
1	2	1	30	10	20
2	6	4	44	16	28
3	8	9	52	26	26
4	10	9	58	26	32

```
mean(tauh)
```

```
[1] 26.5
```

(a) The matches

The banks at $X = 2, 6, 8, 10$ match the controls at 1, 4, 9, 9, giving $\hat{\tau}_i = 20, 28, 26, 32$ and $\widehat{ATT} = 26.5$.

The control at $X = 9$ is **used twice**; the one at $X = 30$ is **never used**. Matching silently discards controls unlike any treated unit — it is a form of trimming. The *ATT* does not need that control, and using it would require extrapolating from $X = 30$ down to the treated range.

(b) Reuse

```
K <- sapply(seq_along(xC), function(j) sum(m == j))
rbind(X_j = xC, K_j = K)
```

```
      [,1] [,2] [,3] [,4]
X_j      1    4    9   30
K_j      1    1    2    0
```

```
sum(K * (K - 1))
```

```
[1] 2
```

$K = (1, 1, 2, 0)$ and $\sum_j K_j(K_j - 1) = 2$.

A reused control puts its own noise ε_j into every comparison that uses it, so those comparisons are **correlated**. The variance penalty counts the *pairs* of treated units sharing a control, which is $K_j(K_j - 1)$. A control used once creates no pair and no penalty. By contrast $\sum_j K_j = N_T M$ whatever the reuse pattern, so it carries no information about dependence.

(c) The Abadie–Imbens standard error

```
NT <- length(xT); M <- 1
sig2 <- ifelse(K > 1, 50, 0) # only reused controls enter term (b)
termA <- sum((tau_h - mean(tau_h))^2) / NT
termB <- sum(K * (K - 1) / M^2 * sig2) / NT
c(termA = termA, termB = termB, sigma2 = termA + termB,
  SE = sqrt((termA + termB) / NT), SE_termA_only = sqrt(termA / NT),
  inflation = sqrt((termA + termB) / termA))
```

termA	termB	sigma2	SE	SE_termA_only
18.750000	25.000000	43.750000	3.307189	2.165064
inflation				
1.527525				

Term (a) is 18.75 and term (b) is $\frac{1}{4} \times 2 \times 50 = 25$, so $\hat{\sigma}_{ATT}^2 = 43.75$ and $SE = 3.31$. Term (a) alone gives 2.17. **Reuse inflates the standard error by a factor of 1.53** — a standard error that ignores it is a third too small.

(d) Bias correction on the matched controls

```
used <- sort(unique(m)) # distinct matched controls
fit_m <- lm(yC[used] ~ xC[used]); coef(fit_m)
```

```
(Intercept)      xC[used]
           8           2
```

```
b_m <- coef(fit_m)[[2]]
bc_m <- tauh - b_m * (xT - xC[m])
data.frame(X_i = xT, gap = xT - xC[m], correction = b_m * (xT - xC[m]),
           tau_i = tauh, tau_BC = bc_m)
```

	X_i	gap	correction	tau_i	tau_BC
1	2	1	2	20	18
2	6	2	4	28	24
3	8	-1	-2	26	28
4	10	1	2	32	30

```
mean(bc_m)
```

```
[1] 25
```

The three matched controls lie exactly on $\hat{\mu}^0(X) = 8 + 2X$. Correcting gives $\hat{\tau}_i^{BC} = 18, 24, 28, 30$ and $\widehat{ATT}^{BC} = 25.0$, down from 26.5.

The largest correction, 4, goes to the bank at $X = 6$: its match is **two** units away (the others are one), and $\hat{\mu}^0$ has slope 2. Matching bias is a **product** of how bad the match is and how steep μ^0 is; either at zero kills it.

Note the bank at $X = 8$: its correction is *positive*. Its match sits above it, so the raw comparison had understated its effect. The discrepancy is signed.

(e) Refitting on all four controls

```
fit_a <- lm(yC ~ xC); coef(fit_a)
```

```
(Intercept)      xC
18.6498054 -0.5136187
```

```
b_a <- coef(fit_a)[[2]]
bc_a <- tauh - b_a * (xT - xC[m])
c(raw = mean(tauh), BC_matched = mean(bc_m), BC_all = mean(bc_a))
```

	raw	BC_matched	BC_all
	26.50000	25.00000	26.88521

Adding the never-matched control flips the slope to $\hat{\beta} = -0.51$, and $\widehat{ATT}^{BC}$ becomes 26.89. The correction has **reversed sign**: fitted on the matched controls it moved the estimate *down* by 1.5; fitted on all four it moves it *up* by 0.39.

One distant control, at $X = 30$ and far outside the treated range, decides the direction of the correction. The “small” regression is doing extrapolation — exactly what matching was meant to avoid — and its answer depends on which controls you feed it. **Report the raw and bias-corrected estimates side by side, and say which controls the outcome model was fitted on.** A large gap is a diagnostic, not a nuisance.

(f) Two covariates and the distance metric

Raw Euclidean distance is dominated by log assets: a one-standard-deviation gap in assets counts twenty times a one-standard-deviation gap in leverage, so you effectively match on assets alone. **Normalized Euclidean** (divide each covariate by its standard deviation) fixes the scale; **Mahalanobis** also accounts for correlation between them.

The fix is **not free**. If leverage barely moves Y^0 , up-weighting it degrades the matches on the covariate that matters. In the lecture’s training data, Mahalanobis returned 1,315 against a truth of 1,607.5, while Euclidean hit it exactly.

Problem 4 (Propensity scores and weighting)

(a) Why balance is free and unconfoundedness is not

The balancing property is free because $e(X)$ is a function of X . Conditioning on it adds nothing to X , so $E[D | X, e(X)] = E[D | X] = e(X)$; and averaging $E[D | X]$ over X within a value of the score gives $E[D | e(X)] = e(X)$. Both sides equal $e(X)$ by the definition and the law of iterated expectations alone.

In the propensity score theorem, unconfoundedness enters at **one step**: replacing $E[D | Y^1, Y^0, X]$ with $E[D | X]$ inside the iterated expectation. Everything else is bookkeeping.

The implication: a balance table checks the balancing property, which holds automatically for a correctly specified score. **Good balance validates the score specification — never unconfoundedness.** It says nothing about variables you did not put in the score, such as managerial quality.

(b) Equal means, unequal spreads

A near-zero mean difference with a variance ratio of 2.4 means treated units are **more dispersed** in this covariate than controls: selection in **both tails**, with treatment more likely at extreme values, high or low. Add the **squared term** to the logit. You know it worked if the variance ratio falls into $[0.8, 1.25]$ — or at least $[0.5, 2]$ — while the standardized difference stays below 0.1, computed with the denominator fixed at its raw-sample value.

A t -test is the wrong diagnostic because balance is a property of *this sample*, not a hypothesis about a population, and the p -value **confounds imbalance with sample size**: large samples flag trivial imbalances, small ones pass large ones, and dropping observations raises p -values by cutting power rather than by improving balance (Imai, King and Stuart 2008).

(c) IPW by hand

```
firms <- data.frame(cell = c(rep("L", 8), rep("H", 4)),
                     D    = c(1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0))
Y0c <- c(L = 10, H = 20); Y1c <- c(L = 13, H = 28)
firms$Y <- ifelse(firms$D == 1, Y1c[firms$cell], Y0c[firms$cell])
firms$tau <- Y1c[firms$cell] - Y0c[firms$cell]
firms$e <- ave(firms$D, firms$cell) # the true cell share
with(firms, round(c(e_L = e[cell == "L"][1], e_H = e[cell == "H"][1],
                    ATE = mean(tau), ATT = mean(tau[D == 1]),
                    SDO = mean(Y[D == 1]) - mean(Y[D == 0])), 4))
```

e_L	e_H	ATE	ATT	SDO
0.2500	0.7500	4.6667	6.0000	10.5714

$e_L = 0.25$ and $e_H = 0.75$. The effects are 3 and 8, so $ATE = 4.67$ and $ATT = 6.00$, while the SDO is 10.57.

```
Nf <- nrow(firms)
EY1 <- with(firms, sum(D * Y / e) / Nf) # Horvitz--Thompson
EY0 <- with(firms, sum((1 - D) * Y / (1 - e)) / Nf)
c(EY1 = EY1, EY0 = EY0, ATE_IPW = EY1 - EY0,
  sum_w_treated = with(firms, sum(D / e)),
  sum_w_control = with(firms, sum((1 - D) / (1 - e))))
```

EY1	EY0	ATE_IPW	sum_w_treated	sum_w_control
18.000000	13.333333	4.666667	12.000000	12.000000

$\widehat{E}[Y^1] = (2 \cdot 13/0.25 + 3 \cdot 28/0.75)/12 = 18$ and $\widehat{E}[Y^0] = (6 \cdot 10/0.75 + 20/0.25)/12 = 13.33$, so $ATE_{IPW} = 4.67$, **exactly the truth**. Each arm's weights sum to $12 = N$: **each arm is blown up into the full population**.

```

w_odds <- with(firms, e / (1 - e))
EY0_T <- with(firms, sum((1 - D) * w_odds * Y) / sum((1 - D) * w_odds))
c(treated_mean = with(firms, mean(Y[D == 1])), reweighted_controls = EY0_T,
  ATT_IPW = with(firms, mean(Y[D == 1])) - EY0_T,
  sum_odds_weights = with(firms, sum((1 - D) * w_odds)), N_T = sum(firms$D))

```

treated_mean	reweighted_controls	ATT_IPW	sum_odds_weights
22	16	6	5
N_T			
5			

The controls get weights 1/3 (L) and 3 (H), which sum to 5 = N_T . They are rebuilt into **the treated firms' covariate distribution** — two L, three H — giving 16, so $\widehat{ATT}_{IPW} = 22 - 16 = 6$, again exact.

(d) A misestimated score: Horvitz–Thompson against Hájek

```

firms$e_bad <- ifelse(firms$cell == "L", 0.20, 0.80)
HT <- with(firms, sum(D * Y / e_bad) / Nf)
Hajek <- with(firms, sum(D * Y / e_bad) / sum(D / e_bad))
c(sum_weights = with(firms, sum(D / e_bad)), HT = HT, Hajek = Hajek, truth = EY1)

```

sum_weights	HT	Hajek	truth
13.75000	19.58333	17.09091	18.00000

The weights now sum to 13.75, not 12. Horvitz–Thompson returns 19.58; Hájek returns 17.09; the truth is 18.

Hájek is guaranteed to be a weighted average of the observed treated outcomes, so it always lies in $[13, 28]$; Horvitz–Thompson divides by 12 regardless of what the weights actually sum to, and carries no such guarantee. Hájek is closer here, but it is still wrong.

Normalizing fixes the **scale** of the weights, not their **relative sizes**. The bad score makes each L firm count four times an H firm instead of three, so L gets 73% of the weight instead of 67%, pulling the answer toward 13. A misspecified score stays misspecified; Hájek removes only one of its symptoms.

(e) A cell with no controls

With cell V treated throughout, $e(V) = 1$. **The ATE and the ATT are both lost**: V's treated firms have no controls from which to impute Y^0 . **The ATU survives**, because V has no controls and so gets zero weight in it.

Retreating to the *ATT* does not help here. That advice works when overlap fails at $\hat{e} \approx 0$ — regions with controls but no treated units, which the *ATT* ignores. Here it fails at $\hat{e} = 1$, treated units with no controls, and that is exactly the side the *ATT* needs: it requires $e(X) < 1$.

Instead, **trim V and say so**: the estimand becomes the *ATT* (or *ATE*) for firms outside V, and ten of the fifteen treated firms are gone. Alternatively, if the policy question concerns firms without the treatment, estimate the *ATU*, which V does not threaten.

Problem 5 (Regression when treatment effects are heterogeneous)

```
cells <- data.frame(X = c(0, 0, 1, 1), D = c(1, 0, 1, 0),
                    n = c(120, 480, 280, 120), y0 = c(10, 10, 20, 20), eff = c(2, 2, 8, 8))
banks  <- cells[rep(1:4, cells$n), ]
banks$Y <- banks$y0 + banks$D * banks$eff
nrow(banks)
```

```
[1] 1000
```

(a) Three estimands, one coefficient

```
ATE5 <- mean(banks$eff)
ATT5 <- mean(banks$eff[banks$D == 1])
ATU5 <- mean(banks$eff[banks$D == 0])
rho  <- mean(banks$D)
tau_hat <- coef(lm(Y ~ D + X, data = banks))["D"]
c(ATE = ATE5, ATT = ATT5, ATU = ATU5, rho = rho, tau_hat = tau_hat)
```

ATE	ATT	ATU	rho	tau_hat
4.4	6.2	3.2	0.4	4.8

$ATE = 4.4$, $ATT = 6.2$, $ATU = 3.2$, $\rho = 0.4$ — and the canonical model returns $\hat{\tau} = 4.8$, **none of the three**.

The failing assumption is the **constant effect**: the effect is 2 for small banks and 8 for large ones. Unconfoundedness holds by design, since assignment is random within class. **Linearity cannot fail** because X is binary: $E[Y^0 | X]$ takes only two values, and two points always lie on a line, so $\alpha + \gamma X$ fits them exactly. The model is saturated in X .

(b) The variance-weighted average

```
e_x <- tapply(banks$D, banks$X, mean)      # e(X): 0.2 small, 0.7 large
N_x <- tapply(banks$D, banks$X, length)
tau_x <- c(2, 8)
s2 <- e_x * (1 - e_x)
w <- N_x * s2
rbind(e = e_x, sigma2_D = s2, weight = w,
      weight_share = w / sum(w), population_share = N_x / sum(N_x))
```

	0	1
e	0.2000000	0.7000000
sigma2_D	0.1600000	0.2100000
weight	96.0000000	84.0000000
weight_share	0.5333333	0.4666667
population_share	0.6000000	0.4000000

```
sum(w * tau_x) / sum(w)
```

```
[1] 4.8
```

$\sigma_D^2 = 0.2 \times 0.8 = 0.16$ for small banks and $0.7 \times 0.3 = 0.21$ for large ones, so the weights are $600 \times 0.16 = 96$ and $400 \times 0.21 = 84$, and

$$\hat{\tau} = \frac{96 \times 2 + 84 \times 8}{96 + 84} = \frac{864}{180} = 4.8. \checkmark$$

Large banks are over-weighted: 47% of the weight against 40% of the population. Their treatment probability, 0.7, is closer to a coin flip than the small banks' 0.2, so $\text{Var}(D | X)$ is larger there — and OLS learns most where treatment varies most.

(c) Through the proof, and through FWL

```
num_x <- sapply(0:1, function(x) { s <- banks$X == x
  mean(banks$Y[s] * (banks$D[s] - e_x[x + 1])) })
rbind(E_Y_times_Dminus_e = num_x, sigma2_times_tau = s2 * tau_x)
```

	0	1
E_Y_times_Dminus_e	0.32	1.68
sigma2_times_tau	0.32	1.68

```
p_x <- N_x / sum(N_x)
sum(p_x * num_x) / sum(p_x * s2)      # ratio of averages
```

```
[1] 4.8
```

```
Dtilde <- banks$D - fitted(lm(D ~ X, data = banks)) # FWL route
cov(banks$Y, Dtilde) / var(Dtilde)
```

[1] 4.8

Cell by cell, $E[Y(D - e) | X]$ is $0.32 = 0.16 \times 2$ for small banks and $1.68 = 0.21 \times 8$ for large ones — **exactly** $\sigma_D^2(X) \tau(X)$, as the proof's Step 2 says. Averaging over the population gives $0.864/0.18 = 4.8$. The Frisch–Waugh–Lovell ratio $\widehat{Cov}(Y, \tilde{D}) / \widehat{Var}(\tilde{D})$ returns the same 4.8: because X is saturated, $\hat{D} = e(X)$ and $\tilde{D} = D - e(X)$, which is precisely the object in the proof.

(d) Regression adjustment

```
ra <- coef(lm(Y ~ D * X, data = banks))
Xbar <- mean(banks$X)
Xbar1 <- mean(banks$X[banks$D == 1])
c(delta = ra[["D"]], lambda = ra[["D:X"]],
  ATE = ra[["D"]] + ra[["D:X"]] * Xbar, ATT = ra[["D"]] + ra[["D:X"]] * Xbar1)
```

delta	lambda	ATE	ATT
2.0	6.0	4.4	6.2

```
banks$Xc <- banks$X - Xbar
coef(lm(Y ~ D * Xc, data = banks))[["D"]]
```

[1] 4.4

$\hat{\delta} = 2$ and $\hat{\lambda} = 6$. Then $\widehat{ATE} = 2 + 6 \times 0.4 = 4.4$ and $\widehat{ATT} = 2 + 6 \times 0.7 = 6.2$ — **both exact**.

$\hat{\delta}$ alone is the effect at $X = 0$, the small banks' effect of 2, not the *ATE*. Centring X at $\bar{X} = 0.4$ moves the reference point to the population average, and the refitted $\hat{\delta}$ becomes 4.4, the *ATE*. Centring at $\bar{X}_1 = 0.7$ would deliver the *ATT* instead.

(e) Słoczyński's weights

```
w1_solved <- (ATU5 - tau_hat) / (ATU5 - ATT5)
p <- fitted(lm(D ~ X, data = banks))
V1 <- mean((p[banks$D == 1] - mean(p[banks$D == 1]))^2) # population variances
V0 <- mean((p[banks$D == 0] - mean(p[banks$D == 0]))^2)
w1_formula <- (1 - rho) * V0 / (rho * V1 + (1 - rho) * V0)
c(w1_solved = w1_solved, w1_formula = w1_formula, V1 = V1, V0 = V0,
  one_minus_rho = 1 - rho, rho = rho)
```

w1_solved	w1_formula	V1	V0	one_minus_rho
0.5333333	0.5333333	0.0525000	0.0400000	0.6000000
rho				
0.4000000				

Solving $4.8 = w_1 \times 6.2 + (1 - w_1) \times 3.2$ gives $w_1 = 1.6/3 = 0.533$, and the formula, with $V[p \mid D=1] = 0.0525$ and $V[p \mid D=0] = 0.04$, returns the same 0.533 exactly.

w_1 **is below** $1 - \rho = 0.6$ because the score varies more among the treated than among the controls ($0.0525 > 0.04$). With equal variances, w_1 would be exactly $1 - \rho$.

Relative to size, **the treated get the larger weight**: they are 40% of the banks but carry 53% of the weight. That is Słoczyński's headline — *the smaller group gets the larger weight* — so the canonical coefficient leans toward the *ATT*, the effect for the minority group, which is rarely the estimand anyone asked for.